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Cite this article: Vega H, Osenberg CW, Seifert AW, Munk N, Stier AC. 2026 Coral-associated fishes accelerate coral wound healing and photosynthetic recovery. *Biol. Lett.* **22**: 20260177.

<https://doi.org/10.1098/rsbl.2026.0177>

Received: 16 March 2026

Accepted: 28 May 2026

Subject Category:

Marine biology

Subject Areas:

ecology

Keywords:

coral regeneration, mutualism, damselfish, *Pocillopora*, wound healing, coral resilience, *Dascyllus flavicaudus*

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Supplementary material is available online at

<https://doi.org/10.6084/m9.figshare.c.8531212>.

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Coral-associated fishes accelerate coral wound healing and photosynthetic recovery

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Coral resilience is increasingly recognized as the product of complex interactions between corals, their endosymbiotic algae and external symbionts such as fishes and invertebrates. While previous studies have documented how coral-associated fishes influence coral growth and thermal tolerance, their role in supporting tissue repair and regeneration following physical injury remains poorly understood. Here, we experimentally test whether coral-associated fishes facilitate wound healing in the coral *Pocillopora* spp. We imposed small and large wounds and monitored healing in the presence and absence of coral-dwelling damselfishes (*Dascyllus flavicaudus*). Corals exposed to fishes exhibited faster wound closure and higher photosynthetic efficiency in recently regenerated tissue, particularly for larger wounds. These findings suggest that coral-associated fishes promote the energetic and microenvironmental conditions necessary for tissue repair. Our results extend the emerging understanding of coral holobiont resilience, highlighting the importance of external symbionts in sustaining coral recovery from physical disturbances.

1. Introduction

Coral reefs are increasingly understood as complex mutualistic systems not only defined by the well-known partnership between corals and their algal symbionts, but also shaped by a broader network of external associations with fishes and invertebrates. Many of these coral-associated fishes and invertebrates live in an intimate relationship with corals, influencing the physical and chemical environments directly surrounding coral tissues [1]. These external partners have been shown to affect nutrient cycling [2], predator–prey dynamics [3], sediment removal [4,5], parasite loads [6] and water flow [7], collectively affecting coral physiology, growth and survival [2,8,9]. As research on these external symbioses grows, it is becoming clear that the resilience of reef-building corals may depend as much on these ecological partnerships as on their internal symbioses (as reviewed in [10]).

Despite this growing appreciation, much of the focus to date has centred on how these associations influence coral responses to large-scale, chronic stressors such as thermal bleaching and ocean acidification [2,11–13]. However, corals also routinely face localized, physical injury from predation, storms,

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disease and human activity [14]. The ability to heal tissue following injury is a fundamental component of coral resilience; for example, if left unhealed, a wound can be colonized by macroalgae or pathogens, potentially threatening the survival of the entire colony [15–17]. Yet wound healing and tissue regeneration impose substantial energetic costs that compete with other metabolic demands: damaged corals exhibit reduced growth rates, compromised reproductive output and depleted lipid reserves, particularly in branching species like *Pocillopora* [18,19]. While we know that intrinsic factors such as wound size, energetic state and algal symbiont density influence healing rates [14], far less is known about how the surrounding biotic environment may shape these repair processes. In particular, the potential role of coral-associated fishes to facilitate or constrain wound healing remains virtually unexplored (but see [20] for a notable exception).

Multiple mechanisms suggest that some coral-associated fishes may contribute to wound repair. By excreting ammonium-rich waste, these fishes provide nitrogen that can be readily assimilated by algal symbionts [2,9], potentially enhancing photosynthetic productivity and fueling the energetically demanding process of tissue regeneration. The effects of egested solid waste and its associated microbiome on coral physiology are, however, less well resolved than this ammonium pathway, and coral-associated fishes are not uniformly beneficial—some species inflict minor wounds during normal activity (for example, *Gobiodon* gobies clearing space for egg deposition). Some species of coral-associated fishes, such as *Dascyllus* damselfishes, exhibit territoriality that can also reduce exposure to corallivores and other physical disturbances that could disrupt wound closure [21,22]. Furthermore, their constant movement within coral branches enhances localized water flow and oxygenation, especially at night when respiration can drive hypoxia inside branching coral colonies [7,23,24]. This additional oxygen delivery could help sustain the high metabolic demands of wound healing under conditions that would otherwise limit tissue repair. These distinct pathways suggest that coral-associated fishes could modify the physical and energetic microenvironment in ways that promote regeneration, yet their role in this process has not been empirically tested.

Here, we directly test whether coral-associated fishes facilitate wound healing in their hosts. We hypothesized that coral-associated damselfish enhance both the rate of wound healing and the physiological recovery of host corals. Using colonies of *Pocillopora* spp., we inflicted small and large wounds and monitored healing trajectories in the presence and absence of resident damselfishes. This experiment provides a direct test of how external symbiotic partners contribute to coral regeneration, and more broadly, illuminates the functional importance of coral-associated biodiversity in promoting resilience to physical disturbance.

2. Material and methods

(a) Study site and system

We conducted this experiment in August 2024 at the Richard B. Gump South Pacific Research Station (17°30' S, 149°50' W) on the northern shore of Mo'orea, French Polynesia. The site features a mosaic of shallow patch reefs consisting of, among others, the branching coral *Pocillopora*. Species-level identification of *Pocillopora* could not be resolved visually or genetically, and we therefore refer to it by genus. *Pocillopora* colonies provide structural habitat for a range of coral-associated species, including trapeziid crabs and damselfishes, which can influence coral physiology and resilience [1,2,12,22]. Among this community of exosymbionts is the yellowtail damselfish (*Dascyllus flavicaudus*), one of the most abundant fish species occupying *Pocillopora*. We therefore quantified the effect of *D. flavicaudus* on wound regeneration and photosynthetic efficiency in *Pocillopora* spp.

(b) Specimen collection and experimental setup

We collected 72 coral fragments from six mature *Pocillopora* spp. colonies (>20 cm in diameter) without damselfish from a reef in the lagoon on the north shore. Twelve fragments were collected from each of the six colonies, and parent ID was not recorded. Fragment morphology was standardized, with each fragment representing a single branch of uniform thickness (3 cm) (supplementary material, figure S1). From the same reef, we also collected 18 adult *Dascyllus flavicaudus* (avg: 5.52 ± 0.70 g) using clove oil and hand nets. We transported corals and fish separately in seawater-filled coolers to a laboratory, where we cut all fragments to a uniform length of 4 cm using a Gryphon C-40 Diamond Band Saw. We then closed the fragmentation wound on each coral by gluing the exposed skeleton onto a ceramic frag plug (Oceans Wonders) using cyanoacrylate gel. We randomly distributed 12 fragments across each of six 95 l flow-through aquaria (60 cm × 35 cm × 45 cm) supplied with unfiltered seawater at a rate of 6 l h^{-1} (supplementary material, figure S2). We then acclimated corals and fish to ambient light and temperature conditions for 3 days. During acclimation, we held fish separate from corals and fish received 2 g of TetraMarine Saltwater Flakes (Tetra US, Blacksburg, VA, USA) three times per day (morning, midday and evening).

Following a 3-day acclimation period, fragments exhibited full tissue regrowth along fragmentation wound margins. We randomly assigned four fragments in each tank to one of three treatments: control (no wound), small wound (avg: $1.48 \pm 0.398 \text{ cm}^2$), or large wound (avg: $3.82 \pm 1.08 \text{ cm}^2$). Using an airbrush and a rectangular stencil, we created a single standardized wound in the centre of the flattest face on experimental fragments. To standardize the magnitude of fish presence across experimental treatments, we used a microbalance to weigh each individual *D. flavicaudus*. We then randomly assigned six individuals, totaling 36 g of biomass, to half of the tanks ($n = 3$). Tanks without fish served as controls.

During the experiment, we fed all tanks—regardless of fish presence—2 g of TetraMarine Saltwater Flakes (Tetra US, Blacksburg, VA, USA) three times per day (morning, midday and evening). After 10 minutes, we siphoned out all uneaten food, including any that had settled on fragments. We scrubbed tanks and coral plugs daily to prevent algal overgrowth.

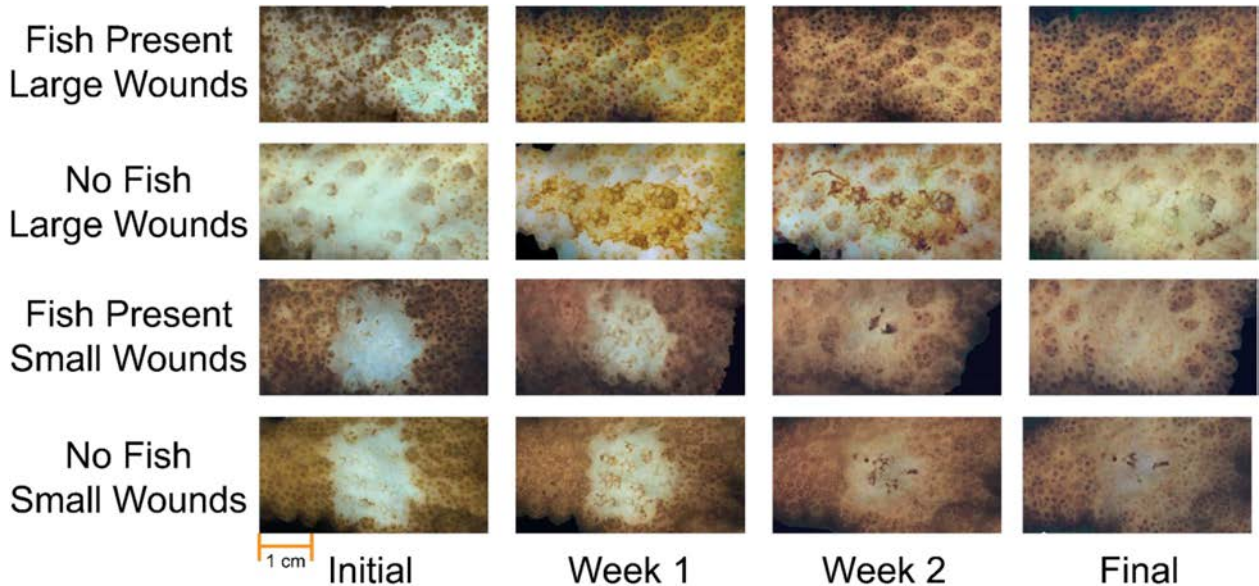


Figure 1. A time series of wound healing by treatment. One coral from each experimental treatment is shown at four time points: initial wounding and one, two and three weeks post-wounding, when the experiment ended. Corals held without fish exhibited less tissue regeneration compared to corals with fish. At three weeks, wounded corals without fish remained only partially healed, while corals with fish exhibited near or complete healing: the small wound with fish is 98% healed, and the large wound with fish is fully healed.

(c) Monitoring wound healing, growth and photosynthetic performance

To assess wound healing and tissue regeneration, we photographed all corals with an Olympus TG-6 digital camera and ruler scale bar immediately post-wounding. We then randomly selected one fragment from each wounding treatment per tank for daily photography (figure 1). On day 21, we photographed all wounds again. Using the initial images, we measured the initial wound size (cm^2) using ImageJ. Using the final images, we quantified the wound area that remained unhealed, which we determined by the absence of visible tissue, regardless of polyp formation. We then calculated the healing rate ($\text{cm}^2 \text{d}^{-1}$) as the change in wound size (initial – final) divided by the duration of the experiment: 21 days. A limitation of this method is that it underestimates the healing rate for any wounds that closed before the end of the experiment. One fragment (no-fish, small-wound treatment) exhibited net tissue loss rather than healing: its wound area increased by 105% over the 21-day experiment (from 1.02 to 2.09 cm^2 ; healing rate = $-0.051 \text{ cm}^2 \text{d}^{-1}$). Because an expanding, partially necrotic wound is a qualitatively distinct outcome from slow healing, this fragment was excluded from the healing-rate analyses ($n = 47$ fragments retained for these models).

To assess coral growth, we measured the buoyant weight of each coral at the start (day 0) and end (day 21) of the experiment [25]. At the end of the experiment, we also measured the surface area of each fragment by wax dipping [26]. For each wounded fragment, the initial wound size was subtracted from the total surface area measurements to provide an accurate measure of the initial surface area of live coral. We calculated coral growth ($\text{mg cm}^{-2} \text{d}^{-1}$) as the area-adjusted calcification rate: (final buoyant weight – initial buoyant weight)/((surface area – initial wound area) $\times$ 21).

To assess coral physiological status on day 21, we measured dark-adapted photosynthetic efficiency (F_v/F_m) using pulse-amplitude modulated fluorometry (Diving-PAM, Walz, Germany) [27]. For corals without wounds, measurements were taken near the centre face of the fragment in the same region where we made wounds in the other corals; for wounded corals, measurements were taken near the centre of the wound. Throughout the healing process, turf algae colonized open wounds. By day 21, algae persisted on the wounds of eight corals. We therefore carefully removed the algae with tweezers before taking PAM measurements. Of these eight corals, seven were in tanks without fish and six had large wounds.

(d) Statistical analyses

We performed all statistical analyses in R (v4.3.1). To assess wound healing rate ($\text{cm}^2 \text{d}^{-1}$), we fitted linear mixed-effects models—with wound healing rate as the response variable—using the lme4 package [28]. Fixed effects included fish presence (present or absent), wound size (small, large) and their interaction, with tank ID included as a random effect to account for non-independence.

To analyse fish and wound effects on coral skeletal growth, we used an ANCOVA-style approach where we included $\log(\text{Initial Mass})$ as a covariate in a full model with treatment effects. This approach simultaneously estimates the allometric coefficient (b) while testing for treatment effects, avoiding the bias that would occur if b were estimated from pooled data ignoring treatments. We fitted a linear mixed-effects model with $\log(\text{Final Mass})$ as the response variable: $\log(\text{Final Mass}) = \alpha + b \times \log(\text{Initial Mass}) + \text{Fish} + \text{Wound} + \text{Fish} \times \text{Wound} + \text{Tank}$, where Fish (present or absent) and Wound (none, small, large) were fixed effects,

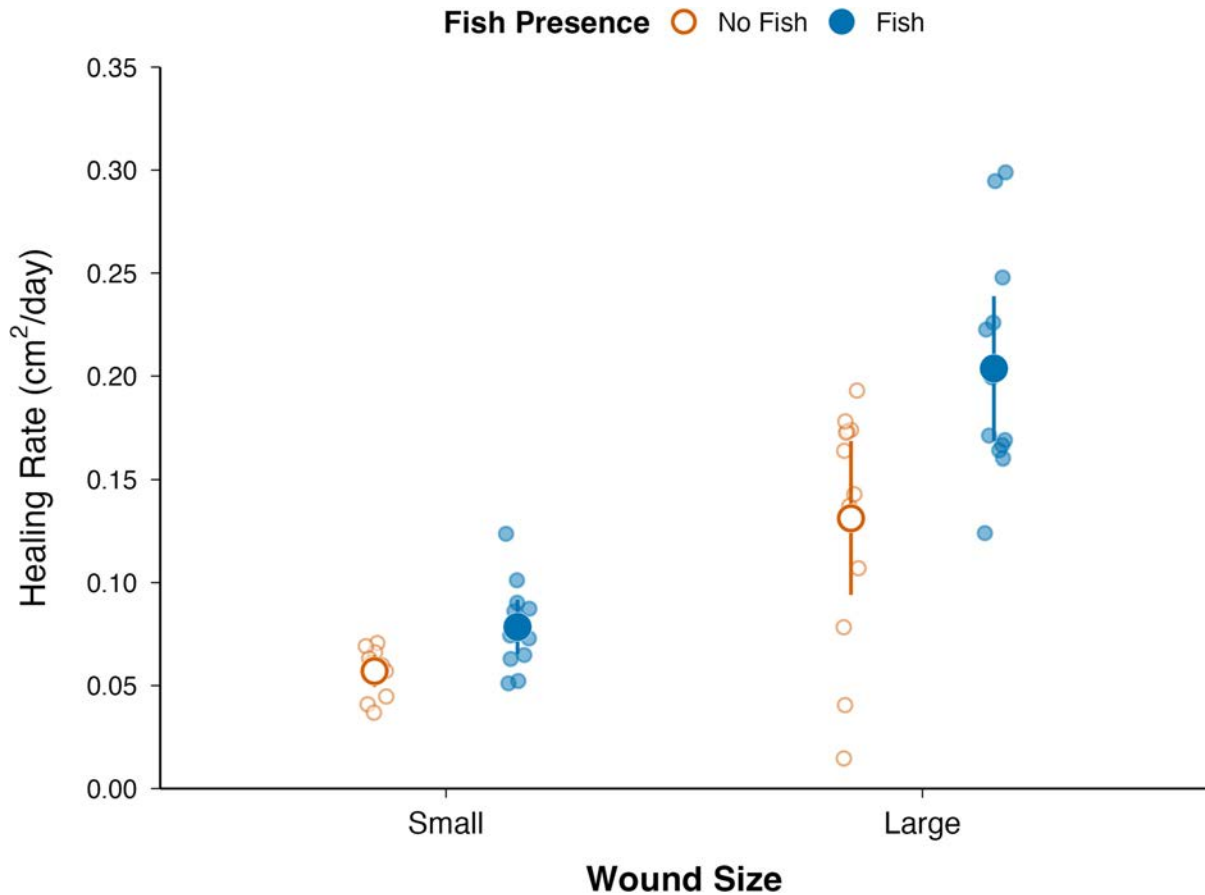


Figure 2. Damselfish presence accelerated coral wound healing, particularly for large wounds. Wound healing rate ($\text{cm}^2 \text{d}^{-1}$) in fragments of *Pocillopora* as a function of wound size and fish presence. Small circles represent individual coral fragments ($n = 8$ per treatment); large circles indicate treatment means with vertical bars showing 95% confidence intervals. Corals with resident damselfish (blue, filled circles) exhibited higher healing rates than fish-absent controls (orange, open circles) across both wound size treatments. This effect was most pronounced in corals with large wounds, where corals with fish healed at approximately twice the rate of corals without fish.

and Tank was included as a random intercept to account for non-independence of corals within the same tank. We then used maximum likelihood estimation and tested the significance of treatment effects using likelihood ratio tests comparing nested models fitted by maximum likelihood (ML). The estimated allometric coefficient was $b = 0.990$ (95% CI: 0.986–0.994), indicating near-isometric growth. For visualization, we calculated size-corrected growth using the b -value estimated from the full model: Scaled Growth = $\log(\text{Final Mass}/\text{Initial Mass}^b)$. We also calculated the absolute growth rate as mass change per unit surface area per day: Growth Rate ($\text{mg cm}^{-2} \text{d}^{-1}$) = $(\text{Final Mass} - \text{Initial Mass})/(\text{Total Surface Area} - \text{Wound Area})/21 \text{ d}$. This metric represents the mass of calcium carbonate deposited per cm^2 of coral tissue per day.

In preparation to analyse the relationship between Fv/Fm and fish presence, we first evaluated the relationship between fish presence and algal persistence on wounds. We used Fisher's exact test on the 2×2 contingency of fish treatment and algae persistence. We then examined variation in Fv/Fm two separate times: once excluding fragments with algal persistence and another time including fragments with algal persistence. We used linear mixed-effects models with fish presence, wound size and their interaction as fixed effects and tank ID as a random effect. For all models, we assumed normal error distributions and homoscedastic variances and evaluated those assumptions using diagnostic residual plots. To assess the significance of fixed effects and their interactions, we used likelihood-ratio tests comparing nested models fitted by ML, using the `anova()` function from the `car` package [29]. We report chi-squared statistics and associated p -values for all terms.

3. Results and discussion

The effect of coral-associated damselfishes on wound healing depended on wound size (fish $\times$ wound interaction: $\chi^2 = 4.416$, $p = 0.036$, figure 2). Fish presence increased healing rates for both wound sizes, with a larger absolute effect for large wounds. For small wounds, fish increased healing rates by 38%, from $0.057 \text{ cm}^2 \text{d}^{-1}$ (95% CI: 0.049–0.065) without fish to $0.079 \text{ cm}^2 \text{d}^{-1}$ (95% CI: 0.065–0.092) with fish. For large wounds, fish increased healing rates by 55%, from $0.131 \text{ cm}^2 \text{d}^{-1}$ (95% CI: 0.094–0.168) without fish to $0.204 \text{ cm}^2 \text{d}^{-1}$ (95% CI: 0.169–0.239) with fish. The absolute effect of fish was 3.3 times greater for large wounds ($0.073 \text{ cm}^2 \text{d}^{-1}$) than for small wounds ($0.022 \text{ cm}^2 \text{d}^{-1}$). Notably, the majority of fish treatment fragments (16/24) had fully closed by day 21, compared to (9/23) without fish. Because a fixed 21-day denominator underestimates the rate for wounds that closed earlier, the fish effect on healing is probably greater than estimated here.

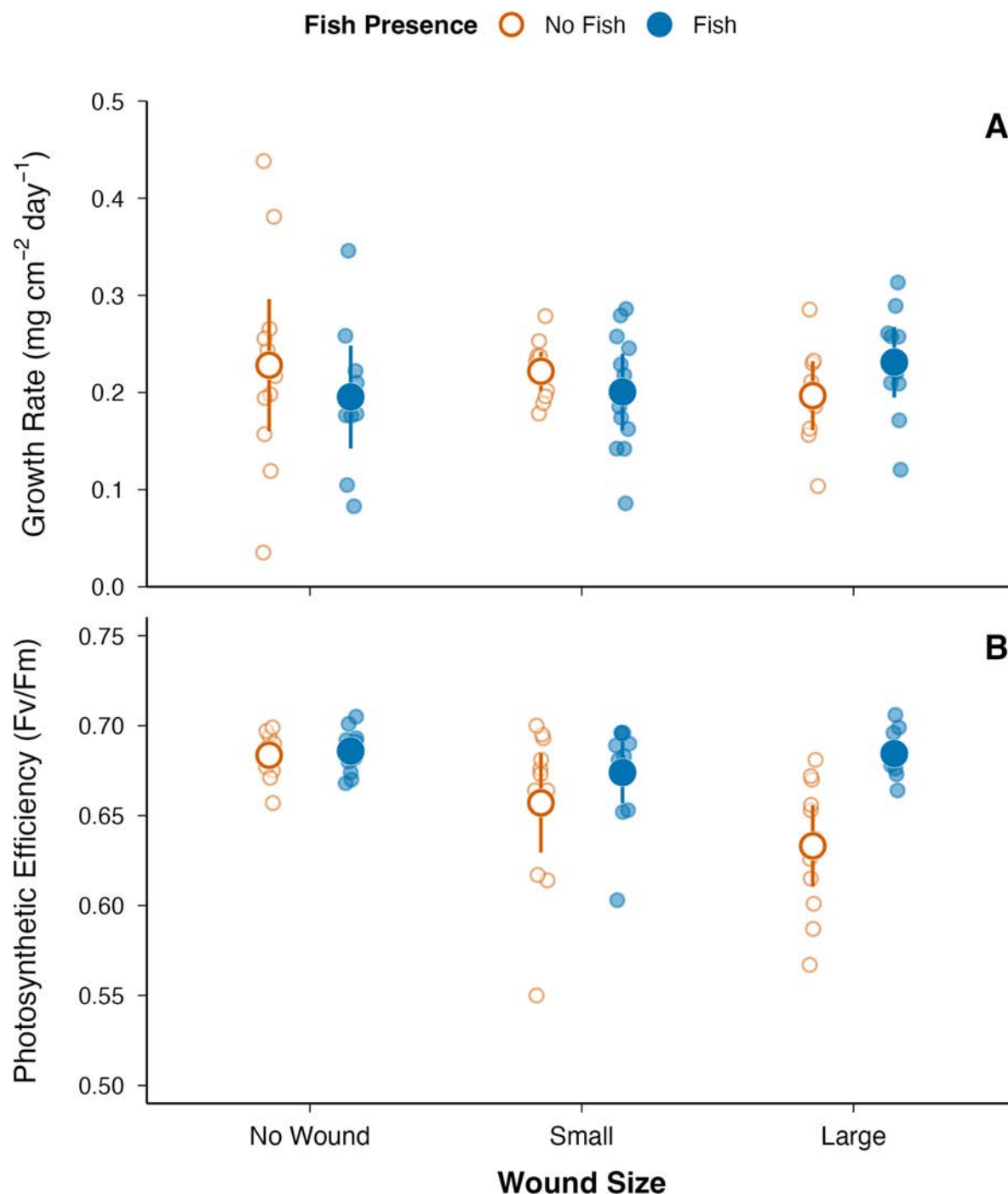


Figure 3. Effects of damselfish on coral growth rate and photosynthetic efficiency across wound treatments. (A) Coral skeletal growth rate (mg d⁻¹ cm⁻²) and (B) maximum quantum yield of photosystem II (F_v/F_m) measured in fragments of *Pocillopora* exposed to three wound size treatments (no wound, small, large) in the presence (blue, filled circles) or absence (orange, open circles) of *Dascyllus* damselfish. Small circles represent individual coral fragments; large circles indicate treatment means $\pm$ 95% confidence intervals. (A) Growth rates were similar across all treatments. (B) Fish presence had minimal effect on photosynthetic efficiency in unwounded corals, but enhanced F_v/F_m in wounded corals, with the largest difference observed in corals with large wounds. Corals without fish showed reduced F_v/F_m in wounded treatments, particularly in the large wound treatment, where corals without fish exhibited the lowest photosynthetic efficiency.

During the healing process, turf algae frequently colonized wounds. By day 21, algae persisted on eight corals and was biased towards tanks without fish (7/8). All three tanks without fish contained at least one fragment with algal persistence, and fragments with no fish were significantly more likely to exhibit algal persistence ($p = 0.048$, odds ratio = 0.11). Without fish, fragments were roughly nine times more likely to exhibit algal persistence.

Algal persistence did not significantly suppress F_v/F_m ($\chi^2 = 6.108$, $p = 0.106$), although the depression in average F_v/F_m between fragments with algal persistence (0.609 (95% CI: 0.582–0.636)) and without the algal persistence (0.677 (95% CI: 0.672–0.683)) is notable. The insignificant result probably reflects low power from a small sample size ($n = 8$) rather than the absence of an effect. Fragments with algal persistence did not meaningfully change our model outputs, and are pooled with all fragments in the following analysis (supplementary material, table S3 and figure S4). We found that the effect of damselfish on F_v/F_m depended on wound size (fish $\times$ wound interaction: $\chi^2 = 10.898$, $p = 0.0043$, figure 3A). In the presence of damselfish, photosynthetic efficiency

was similar across all three wounding treatments: No Wound = 0.686 (95% CI: 0.679–0.693), Small = 0.674 (95% CI: 0.657–0.691) and Large = 0.684 (95% CI: 0.677–0.692). In the absence of fish, however, photosynthetic efficiency declined with increasing wound size: No Wound = 0.684 (95% CI: 0.676–0.691), Small = 0.657 (95% CI: 0.629–0.685) and Large = 0.633 (95% CI: 0.610–0.656). Fish presence had minimal effect on unwounded corals (+0.002 *Fv/Fm* units) but progressively greater effects on wounded corals: +0.017 for small wounds and +0.051 for large wounds. Thus, damselfish presence positively influenced photosynthetic efficiency, and these effects were most pronounced when wounds were larger and metabolic demands associated with wound repair were presumably higher.

In contrast to effects on healing rate and photosynthetic efficiency, and contrary to previous studies [2,9], we observed no significant effects of fish presence, wound size or their interaction on coral growth after controlling for initial coral size (Fish: $\chi^2 = 0.001$, $p = 0.978$; Wound: $\chi^2 = 1.783$, $p = 0.410$; Fish $\times$ Wound: $\chi^2 = 5.224$, $p = 0.073$; figure 3B). Given prior studies demonstrating positive effects of fish on coral growth, and our finding that fish increased photosynthetic efficiency and healing rate, we suggest that the absence of detectable calcification responses may reflect the short experimental duration of our study. It may also reflect the lack of an effect of acute wounding stress on calcification, but this seems improbable given past studies [2,9,30].

These findings highlight the critical contribution of coral-associated fishes to coral resilience, specifically their roles in tissue regeneration and physiological recovery following acute injuries. Although the exact mechanisms remain to be elucidated, nutrient subsidies from damselfish excretion probably enhanced symbiotic algal productivity, providing the additional energy required for wound repair [2,9]. Additionally, turf algae persistence in wounds was significantly more common in corals without fish, suggesting that fish may also facilitate wound repair by mitigating the energetic constraints of competitive exclusion. We did not observe fish grazing on wound turf, but our design cannot fully separate this direct route from an indirect one in which faster healing shortens the window of bare substrate available for algal colonization; both are consistent with the data. All tanks, with and without fish, also received the same supplemental feeding, so the inferred nutrient pathway is additional to—and not confounded with—that shared food source. Consistent with this interpretation, corals without fish also exhibited depressed photosynthetic efficiency; however, this may reflect impaired symbiont performance or reduced tissue availability for symbiont colonization, rather than a direct effect on symbiont physiology alone.

In addition to nutrient benefits, damselfish behavior, such as constant swimming within coral branches, may improve water flow and oxygenation; factors that are particularly important during the nighttime period when the metabolic demands of wound healing are high, and hypoxia risk increases [7,23,24]. During this experiment, we observed *D. flavicaudus* often swimming and sheltering among coral fragments, especially at night. Together, the behavioural and nutrient benefits of coral-associated fishes probably enhance the coral's capacity to regenerate tissue efficiently. Yet it should also be noted that the low replication ($n = 3$), standardized flow, lighting and feeding conditions of our aquaria may modify the effect sizes of the proposed mechanisms relative to field conditions. Additionally, the natural variability in responses to these proposed mechanisms may vary, as the genetic variation in coral fragments and donor colonies was not incorporated into this study.

Our results contribute to the emerging perspective that coral-associated fishes enhance coral holobiont resilience by adding the facilitation of tissue regeneration to the previously established roles of fishes in promoting coral growth and increasing thermal tolerance [22,31]. By demonstrating that damselfishes actively facilitate coral regeneration following injury, we underscore the functional importance of external symbiotic relationships in buffering corals from both chronic stressors and acute physical disturbances. Preserving these ecological partnerships will probably be crucial for sustaining coral regenerative capacity and enhancing long-term reef resilience.

Ethics. All procedures involving coral fragments and damselfish (*Dascyllus flavicaudus*) were conducted following institutional and national guidelines for the care and use of animals in research. The study protocol received ethical approval from the University of California, Santa Barbara Institutional Animal Care and Use Committee (Protocol No. 926.2). Research was completed under permits issued by the Government of French Polynesia (Délégation à la Recherche) and the Haut-Commissariat de la République en Polynésie Française (DTRT). We followed best practices to minimize stress and injury to both corals and fishes throughout the experiments.

Data accessibility. Data and analysis code supporting the paper are available in the Damselfish–Coral Healing Project repository: https://github.com/haydenvega/damselfish_coral_healing_public (v1.0). Cleaned datasets and analysis scripts are licensed under CC BY 4.0. Raw field data are archived within the repository and have been deposited in a long-term archive at Zenodo [32].

Supplementary material is available online [33].

Declaration of AI use. We have not used AI-assisted technologies in creating this article.

Authors' contributions. H.V.: conceptualization, data curation, formal analysis, funding acquisition, investigation, methodology, project administration, resources, visualization, writing—original draft, writing—review and editing; C.W.O.: conceptualization, funding acquisition, project administration, visualization, writing—original draft, writing—review and editing; A.W.S.: formal analysis, methodology, writing—original draft, writing—review and editing; N.M.: conceptualization, formal analysis, funding acquisition, investigation, methodology, project administration, writing—original draft, writing—review and editing; A.C.S.: conceptualization, data curation, formal analysis, funding acquisition, investigation, methodology, project administration, resources, supervision, validation, visualization, writing—original draft, writing—review and editing.

All authors gave final approval for publication and agreed to be held accountable for the work performed therein.

Conflict of interest declaration. We declare we have no competing interests.

Funding. Our project benefited from support from the Keck Foundation and the National Science Foundation (OCE-1851510 and OCE-1851032). This work was also facilitated by resources provided by the NSF-funded Moorea Coral Reef Long-term Ecological research programme funded by OCE-1637396 and previous awards.

Acknowledgements. We thank Kelsey Vaughn, Christian Deneka and Adrian Cheh for their assistance during specimen and data collection. The staff at Gump Station provided invaluable assistance to field and wet laboratory operations.

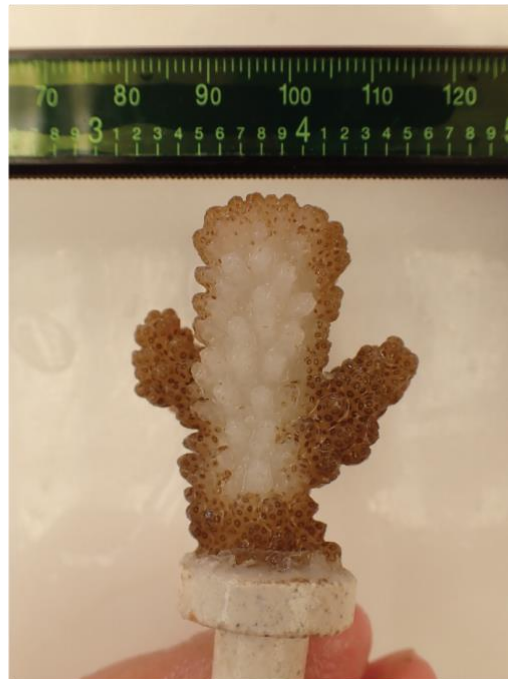
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Supplementary Figures and Tables (S1 - S4) for Vega et al. 2026: Coral-Associated Fishes Accelerate Coral Wound Healing and Photosynthetic Recovery.

Figure S1. (A) A photo of coral of an experimental coral fragment immediately after receiving a large wound. (B) A photo of an experimental coral immediately after receiving a small wound. Fragments are roughly 2cm x 4cm. Rectangular wounds were placed in the center of the flattest face present. A scale bar is shown in both images with mm (top) and inches (bottom).

A



B

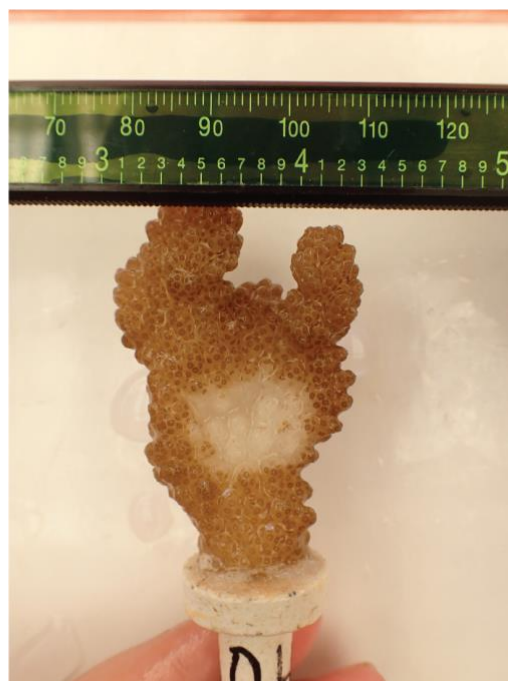


Figure S2. (A) Dimensions of the 95L experimental aquaria. (B) Schematic of coral fragments within experimental aquaria (top-down view); each circle represents one coral fragment.

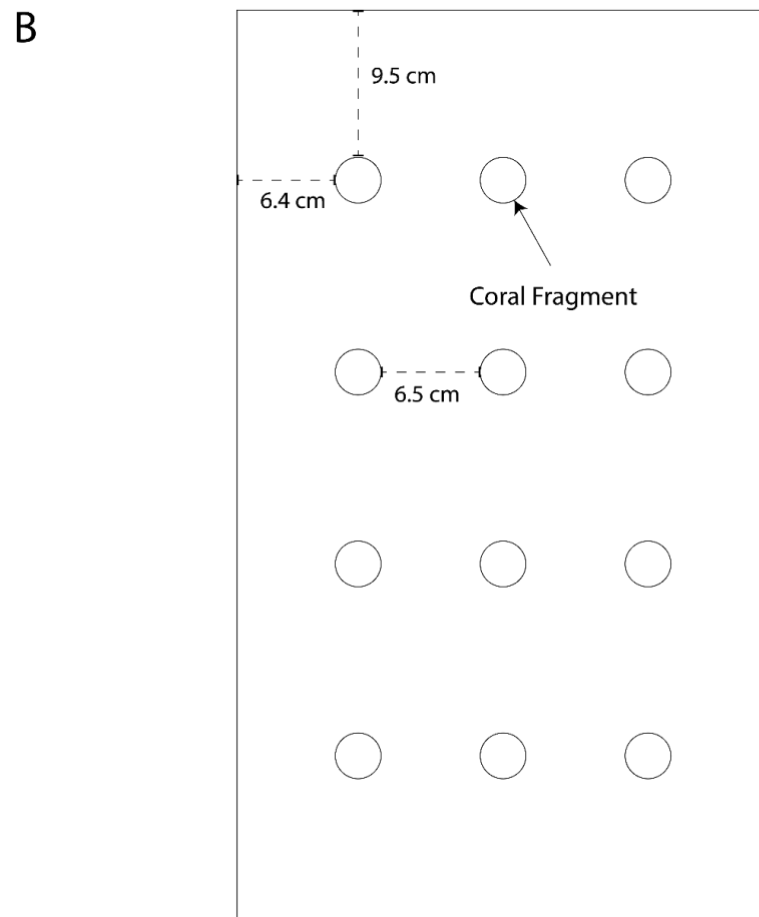
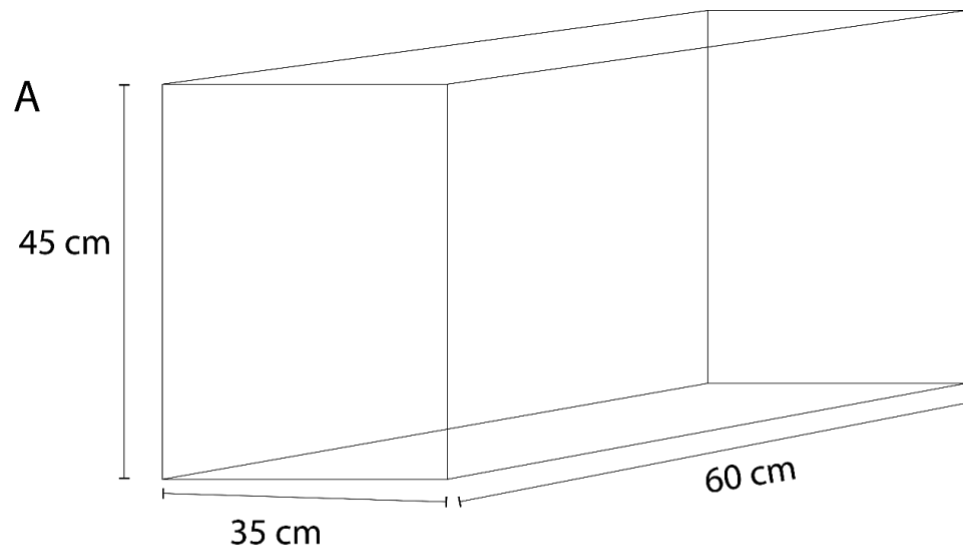


Table S3. (A) Likelihood Ratio Test outputs for mixed-effect models (Fv/Fm ~ Fish Presence * Wound + Tank ID) for photosynthetic efficiency. All 72 experimental corals are included in this analysis. (B) Likelihood Ratio Test outputs for mixed-effect models (Fv/Fm ~ Fish Presence * Wound + Tank ID) for photosynthetic efficiency. All eight corals with algal colonization have been excluded, leaving 64 experimental corals. The significance of the interaction between fish and wound is maintained between both groups of corals, as well as the direction of effects of fish and wounds on photosynthetic efficiency.

A	Likelihood Ratio Tests for Effects on Fv/Fm			
	Full Dataset, n = 72			
	Model Comparison	χ^2	df	P-value
	Interaction (Fish × Wound)	10.898	2	0.004
	Fish Effect	9.104	1	0.003
B	Wound Effect	10.806	2	0.005
	Mixed-effects models with tank as random effect			
	Likelihood Ratio Tests for Effects on Fv/Fm			
	Algal Colonization Excluded, n = 64			
	Model Comparison	χ^2	df	P-value
	Interaction (Fish × Wound)	10.2480	2	0.0060
	Fish Effect	0.6633	1	0.4154
	Wound Effect	7.5165	2	0.0233
	Mixed-effects models with tank as random effect			

Figure S4. Effects of damselfish on photosynthetic efficiency across wound treatments—excluding fragments colonized by algae (n=64). Maximum quantum yield of photosystem II (Fv/Fm) measured in fragments of *Pocillopora* exposed to three wound size treatments (No Wound, Small, Large) in the presence (blue, filled circles) or absence (orange, open circles) of *Dascyllus* damselfish. Small circles represent individual coral fragments; large circles indicate treatment means $\pm$ 95% confidence intervals. Fish presence had minimal effect on photosynthetic efficiency in unwounded corals, but enhanced Fv/Fm in wounded corals, with the largest difference observed in corals with large wounds. Corals without fish showed reduced Fv/Fm in wounded treatments, particularly in the large wound treatment where corals without fish exhibited the lowest photosynthetic efficiency. Algal fragment exclusion does not change the direction of relationships observed between Fv/Fm and experimental treatments. Yet, the magnitude of fish effects decreases, particularly for the large wound treatment—to which 6/8 excluded corals belong.

